AYUSH SETU

Unified Academia-Industry Internship and Skill Alignment Portal

Technical Overview and Platform Guide

Version 1.0 | Academic Year 2025-26

1. Executive Summary

AYUSH SETU is a centralized Academia-Industry Collaboration Portal that bridges the gap between skills taught in educational institutions and competencies demanded by employers in the Ayush sector. The platform connects four key personas: Students, Industry Partners, Academicians, and Administrators on a single unified interface powered by real-time data, AI-driven skill assessment, and a verifiable digital competency passport.

2. Platform Personas

Persona	Core Role	Key Features
Student	Skill discovery and applications	Skill assessment, AI quiz, internship browsing, application tracking
Industry	Post internships and hire talent	Post opportunities, review applications, shortlist candidates
Academician	Bridge academia and industry	Post internships, mentor students, track placements
Admin	Platform governance	Verify credentials, manage users, review flagged content

3. Core Features

3.1 Skill Assessment Engine

Students select domains relevant to their target job profile. An AI-generated quiz tests their knowledge across those domains. Results produce a skill profile highlighting strengths, gaps, and improvement recommendations.

3.2 AI Credential Verification

A regex/fuzzy-matching tokenizer validates academic institution names against a corpus of accredited Ayush colleges. Verified accounts receive a badge increasing credibility for applications.

3.3 Internship Marketplace

Industry partners and academicians post internship opportunities tagged by domain, stipend, mode, and duration. Students filter and apply directly. Application status updates in real time.

3.4 Competency Passport

A shareable digital credential aggregating assessment scores, verified institution, work experience, and certifications. Exportable as PDF for internship applications.

3.5 Real-Time Notification Bus

A LocalStorage event bus and CustomEvent dispatcher ensures actions by one persona are immediately reflected across all open sessions without a page refresh.

3.6 Admin Dashboard

Administrators review flagged content, verify credentials, manage user roles, and publish platform-wide announcements.

4. Technical Architecture

Layer	Technology	Purpose
Frontend	React 19 + Vite	Component-based SPA
Styling	Tailwind CSS v4	Utility-first responsive design
State	Zustand + LocalStorage events	Pub/Sub reactive store
Backend	Supabase (PostgreSQL)	Auth, database, realtime
AI Engine	Regex / Fuzzy tokenizer	Institution verification
Hosting	Vercel	Edge-deployed CDN

Three-tier architecture: (1) React 19 SPA presentation layer, (2) Zustand state with localStorage pub/sub, (3) Supabase PostgreSQL with Row-Level Security.

5. Key Process Flows

5.1 Student Onboarding and Skill Assessment

Step 1	Student registers and selects role (Student), enters institution name.
Step 2	AI tokenizer checks institution against Ayush corpus and returns verification status.
Step 3	Student selects target job domain and preferences (mode, salary, location).
Step 4	AI quiz generated for selected domains; student submits answers.
Step 5	System scores answers and generates skill profile with gap analysis.
Step 6	Competency Passport created; student can now apply for internships.

5.2 Internship Posting and Application Flow

Step 1	Industry or Academician logs in and navigates to Post Internship.
Step 2	Fills in role, domain, stipend, mode, duration, and eligibility criteria.
Step 3	Opportunity published and visible to all students in Internship Marketplace.
Step 4	Student browses filtered listings and clicks Apply Now.
Step 5	Application recorded in real-time store; both parties see updated status.
Step 6	Industry shortlists or rejects; student notified instantly.

5.3 Admin Verification Workflow

Step 1	User submits credentials (institution, role, documents).
Step 2	Admin dashboard flags new submissions for review.
Step 3	Admin approves or rejects; user account status updated.
Step 4	Verified users receive badge; increased trust on the platform.

6. Security and Data Privacy

- Supabase Row-Level Security (RLS) ensures each user can only access their own data.
- OAuth 2.0 / Magic Link authentication - no passwords stored.
- Environment variables (.env.local) used for all API keys - never committed to VCS.
- Input sanitization on all user-submitted text fields.
- Admin-only routes protected by role-based access control (RBAC).

7. Deployment and DevOps

- Hosted on Vercel with automatic CI/CD from the main branch.
- Vite build produces optimized production bundles (tree-shaking, code splitting, lazy routes).
- vercel.json configured for SPA fallback routing.
- Supabase project on free tier with auto-scaling capabilities.

8. Future Roadmap

Phase	Feature	Priority
Phase 1	Mobile-responsive PWA + offline caching	High
Phase 1	Email/SMS notifications for application updates	High
Phase 2	LLM-powered personalized learning path recommendations	Medium

Phase 2	Video interview scheduling integration	Medium
Phase 3	Blockchain-backed verifiable credential issuance	Low
Phase 3	Industry analytics dashboard (hiring trends)	Low